ASIAN INSTITUTE OF MANAGEMENT

Machine Learning 3

Introduction to ML3

Session 1

MSDS 2026

10 April 2026

ASIAN INSTITUTE OF MANAGEMENT

At your service!

Leodegario Lorenzo II

Data Scientist, ACCeSs@AIM

MSDS 2021

Kristine Ann Carandang, OTRP

Data Scientist, ACCeSs@AIM

MSDS 2022

Introduction to ML3

Introduction to ML3

Course Description

This is the third course in the Machine Learning (ML) series, focusing on **deep learning (DL) architectures** and **algorithms**. Students will learn about both fundamental and state-of-the-art architectures such as **Convolutional Neural Networks (CNN)**, **Recurrent Neural Networks (RNN)**, and **Transformers**.

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Course Description

As done in the first two courses, this will be a **hands-on** and **application-heavy module**, looking at **real-world data** and use cases from different fields such as computer vision and natural language processing.

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Course Description

Students will strengthen their ability to **build deep learning models** and **closely link their results** with **business value** or **impact** expected by shareholders.

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Course Description

The most ***recent developments*** in deep learning will also be discussed in class through the **journal scan presentations**.

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Course Description

The **culminating activity** is a presentation of the students' projects that will be judged based on *impact*, *novelty*, and *mastery* of deep learning methodologies.

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Course Learning Outcomes

1. **Design** and **apply** a deep learning pipeline for **computer vision** and **sequence data** tasks.
2. **Explain** the **important concepts** behind various **deep learning architectures**.
3. **Implement** the fundamentals of **CNNs**, **RNNs**, **Transformers**, and other architectures.
4. **Present** the results of recent or high-impact peer reviewed **journal/conference papers**.
5. **Propose novel** machine learning techniques or applications, create technical writeups, and **communicate the result to the public**.

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Course Schedule

Sessions 1 to 2 – Course Introduction and Review of ML Concepts

Sessions 3 to 6 – Convolutional Neural Networks

Sessions 7 to 10 – Recurrent Neural Networks

Sessions 11 to 12 – Transformers Architecture

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Course Schedule

Sessions 13 to 14 – Journal Scan and Perspectives Beyond ML3

Sessions 15 to 16 – Final Exam

Sessions 17 to 20 – Final Project Consultations and Mock Presentations

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Grading Criteria and Course Deliverables

Class Participation	Post-Class Reflection Contribution to Class Learning Hands-on Activities: Exercises and HW	25%
Individual Reports	Mini-Project	20%
Exam	Final Exam	20%
Presentation	Journal Scan Presentation	10%
Group Project	Final Project Presentation and Report	25%

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Grading Criteria and Course Deliverables

Class Participation 25%

Post-Class Reflection – Reflective questions at the end of each class (via ALICE).

Contribution to Class Learning – Subjective evaluation of meaningful participation

Hands-on Activities – Hands-on coding and problem solving (via GitHub classroom)

Exercises (Individual) and Homeworks (by Group)

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Grading Criteria and Course Deliverables

Individual Report 20%

Mini-Projects (1 for CNN and 1 for RNN/Transformers) – An individual project designed to demonstrate mastery in implementing deep learning concepts, particularly CNNs and RNN/Transformers. Evaluation will be based on both the submitted code and written report, following a prescribed grading rubric.

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Grading Criteria and Course Deliverables

Final Exam 20%

An individual, offline assessment designed to evaluate your mastery of core deep learning concepts. The exam will likely consist of three components: (1) Multiple Choice, (2) Problem Solving, and (3) Essay.

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Grading Criteria and Course Deliverables

Journal Scan Presentation 10%

A 10-min presentation (+5 mins Q&A) of a recent or high-impact a peer-reviewed journal or conference paper of a deep learning methodology. The paper should be within the last 2 years published in top-tier journal/conference.

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Grading Criteria and Course Deliverables

Journal Scan Presentation 10%

General AI/ML – ICLR, NeurIPS, ICML, AISTATS, TPAMI, AAAI, IJCAI

Computer Vision – CVPR, ECCV, ICCV

NLP – ACL, EMNLP, NAACL

Data Mining – TKDE, KDD/SIGKDD

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Grading Criteria and Course Deliverables

Final Project and Report 25%

An 8-min presentation (+5 mins Q&A) and written report (journal submission format) of a deep learning project that demonstrates mastery of deep learning methodologies, novelty in problem formulation, and clear articulation of real-world or business impact. All project topics are subjected to prior approval.

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Learning Resources

Lecture Code and Slides

Exercises and Homeworks

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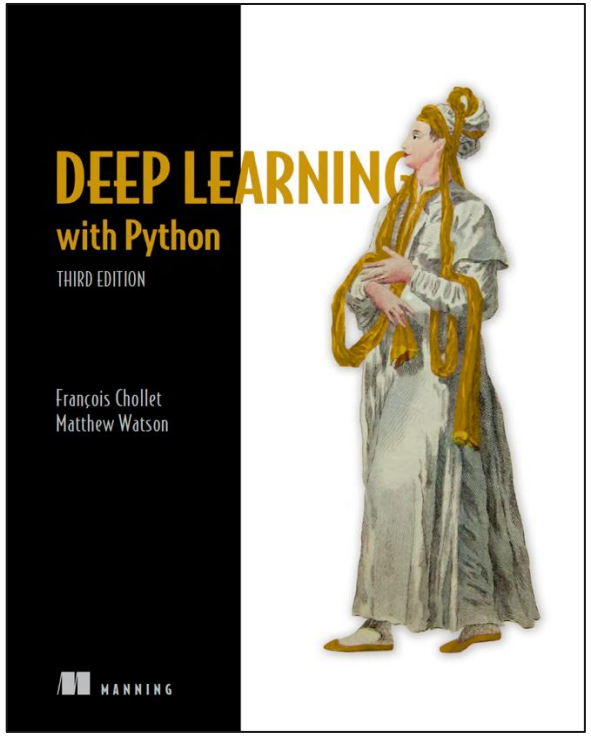
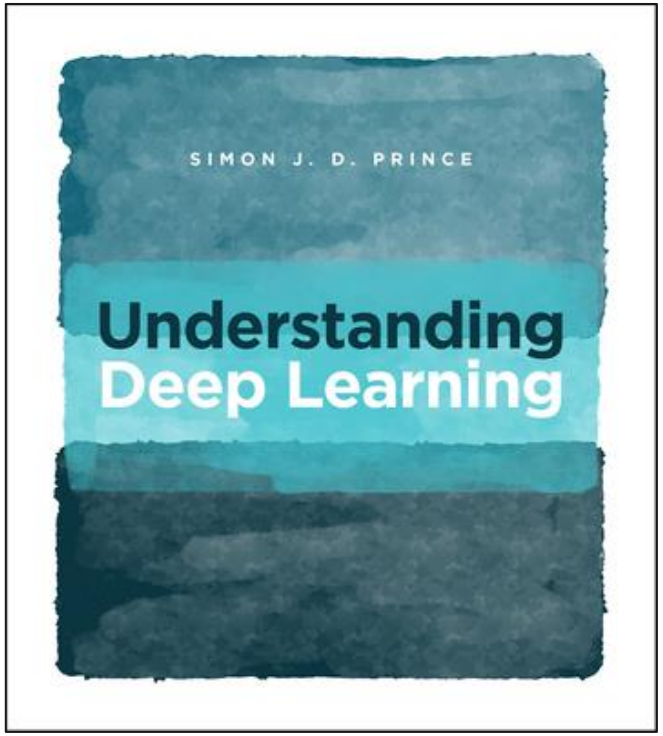
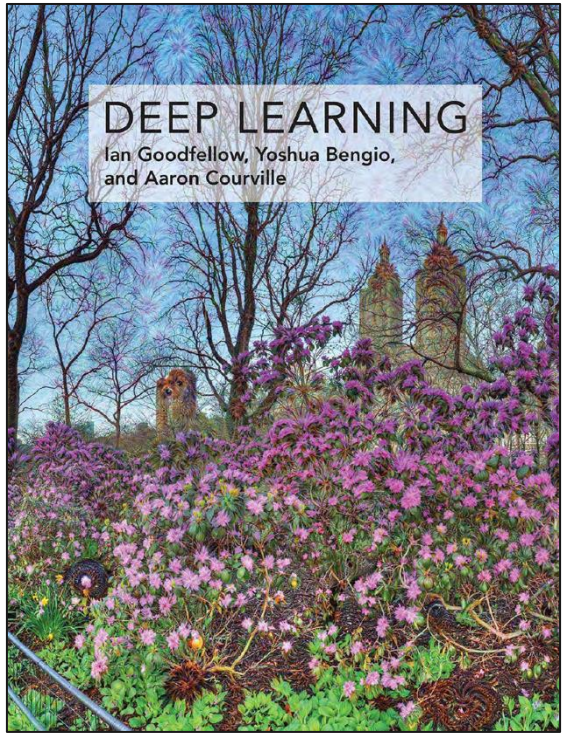
Learning Resources







Introduction to Deep Learning

Keras: Our DL Framework of Choice

TensorFlow

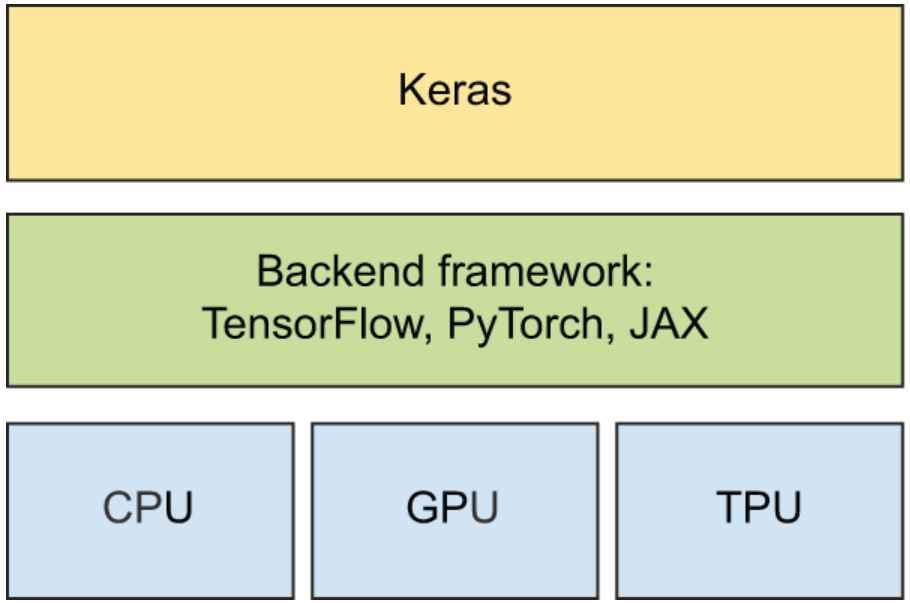
PyTorch

Google JAX

Chollet, F., & Watson, M. (2026). *Deep learning with Python Third Edition*. Manning.

Introduction to Deep Learning

Keras: Our DL Framework of Choice



Deep learning development: layers, models, optimizers, losses, metrics, training loops ...

Tensor manipulation infrastructure: tensors, variables, automatic differentiation, distribution ...

Hardware: execution

Chollet, F., & Watson, M. (2026). *Deep learning with Python Third Edition*. Manning.

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Generative AI Policy

Generative AI technology is considered a **tool or reference** similar to Wikipedia and Stack Overflow. You may use them in the same manner as you use these tools, and as such, must be **cited and acknowledged**. Your submissions should be your intellectual output and **NOT lifted directly** from their output. Passing off their output as your own will be considered as an **academic misconduct** and will be reported to the Office of the Dean for appropriate action.

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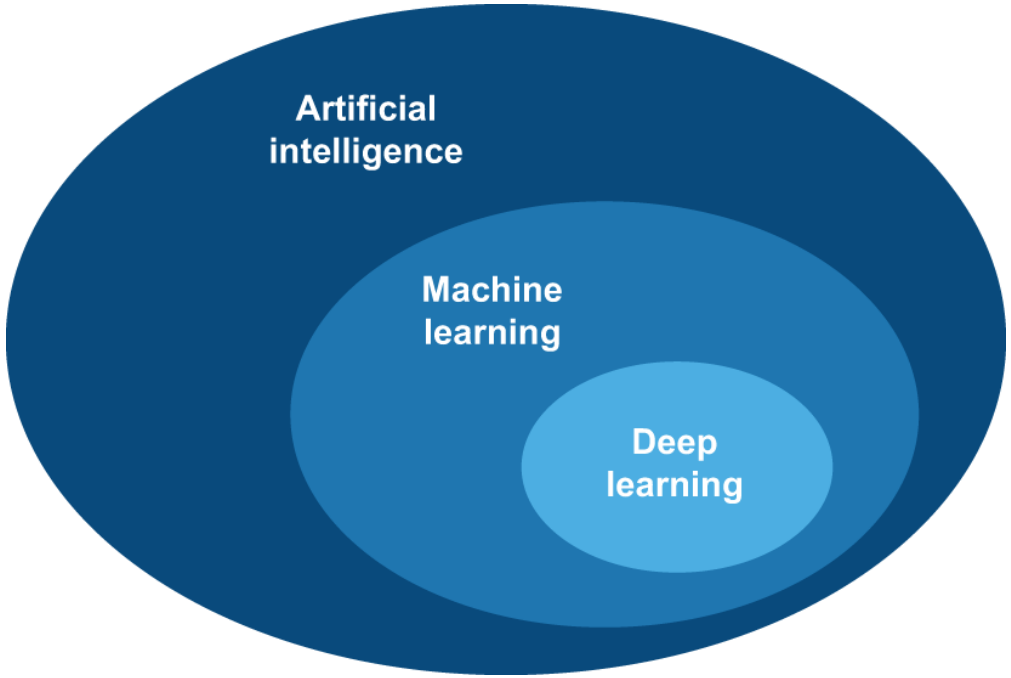
AI vs ML vs DL



Chollet, F., & Watson, M. (2026). *Deep learning with Python Third Edition*. Manning.

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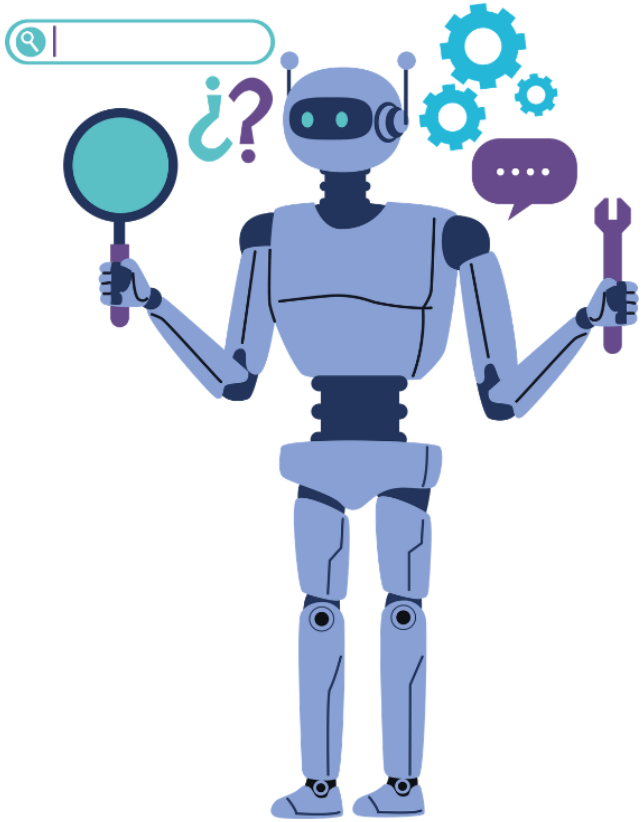
What is Artificial Intelligence?

A branch of computer science that enables machines to **simulate human intelligence**. It can be concisely described as the effort to automate intellectual tasks normally performed by humans.



https://www.tp-link.com/ph/blog/2057/what-is-artificial-intelligence-its-definition-uses-and-types

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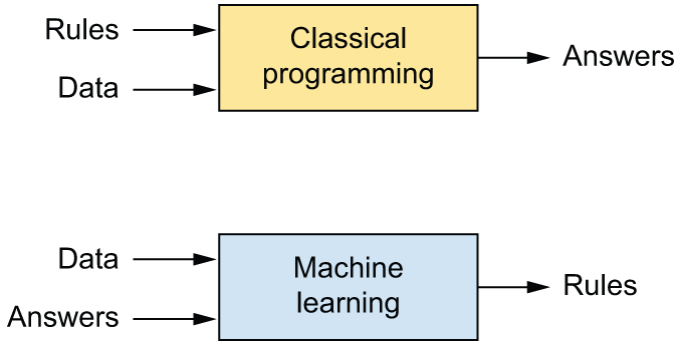
What is Machine Learning?

A subset of artificial intelligence (AI) focused on algorithms that can “learn” the patterns of training data and, subsequently, make accurate inferences about new data. This pattern recognition ability is formed without explicit, hard-coded instructions.



https://www.ibm.com/think/topics/machine-learning

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Key Requirements for Machine Learning

To do machine learning we need three things:

Input data points – a numerical representation of observations used for prediction.

Examples of expected output – the expected output based given input data points.

Error function – way to measure whether the algorithm is doing a good job.

Chollet, F., & Watson, M. (2026). *Deep learning with Python Third Edition*. Manning.

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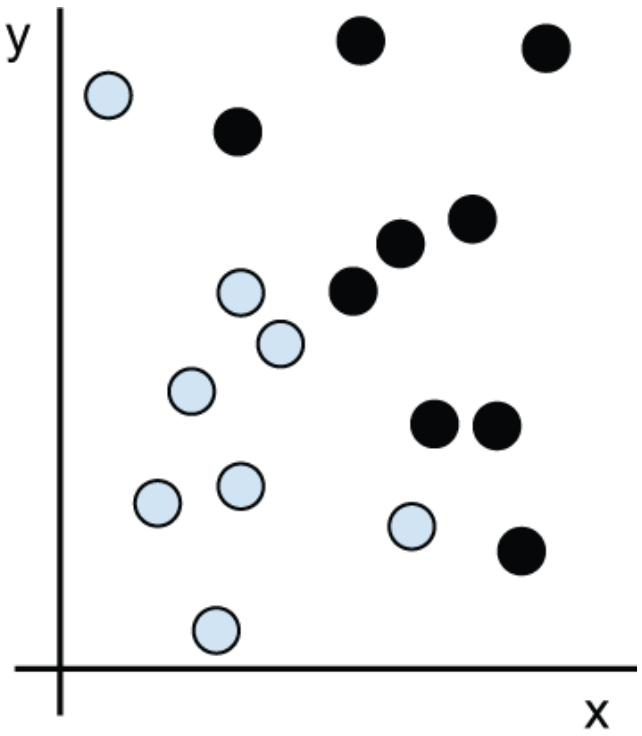
Key Requirements for Machine Learning



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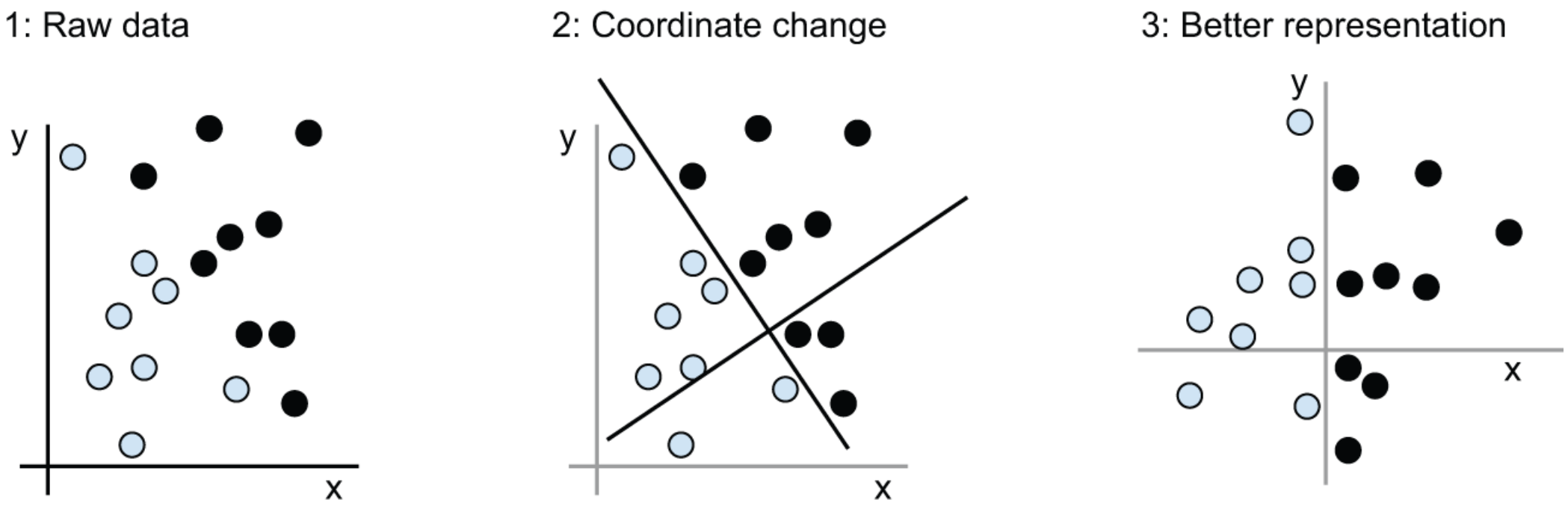
Key Requirements for Machine Learning



Chollet, F., & Watson, M. (2026). *Deep learning with Python Third Edition*. Manning.

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Key Insight in Machine Learning

Machine learning is all about – searching for useful *representations* and *rules* over some input data using guidance from a feedback signal.

Chollet, F., & Watson, M. (2026). *Deep learning with Python Third Edition*. Manning.

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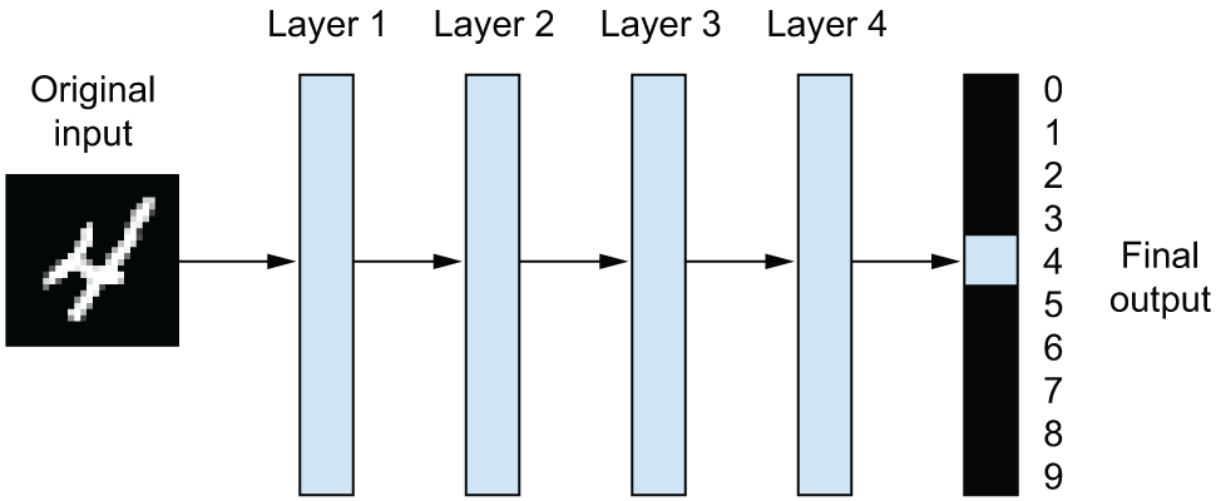
What makes deep learning "different"?

Deep learning is a subfield of machine learning that emphasizes learning *successive layers* of increasingly meaningful representations.



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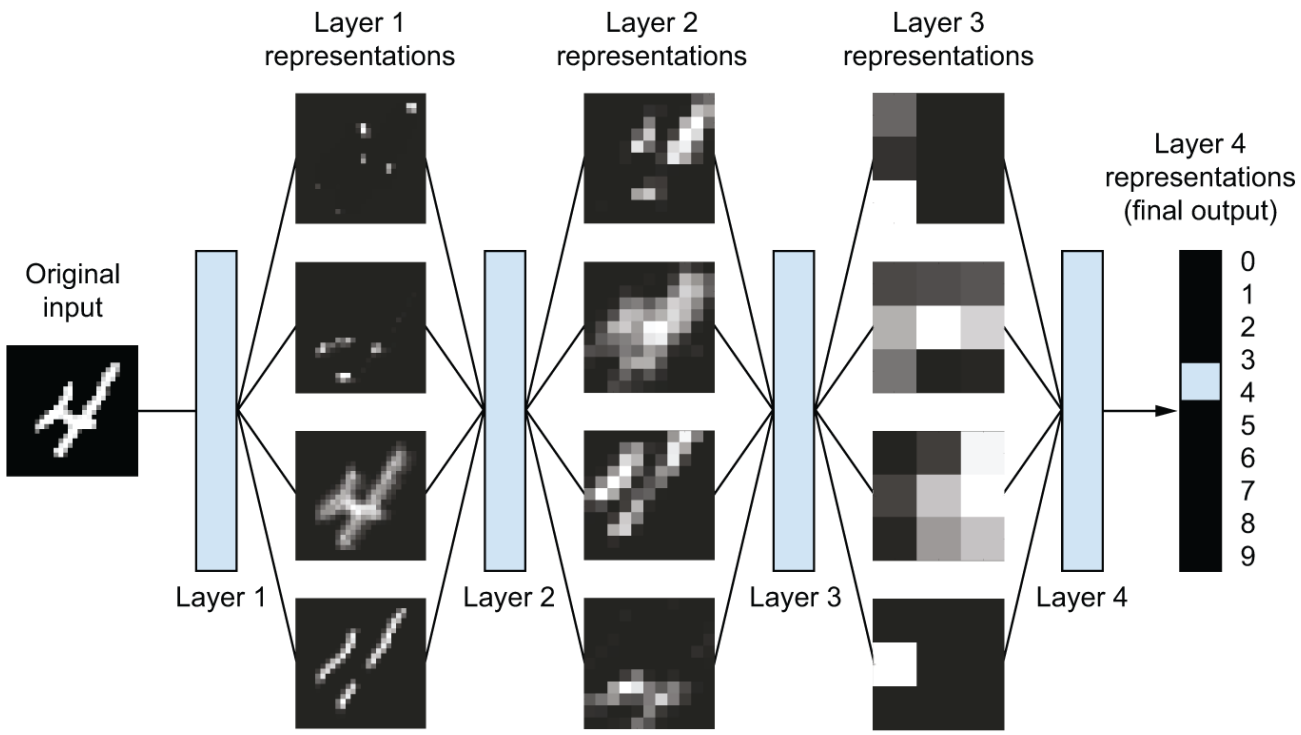
What makes deep learning "different"?



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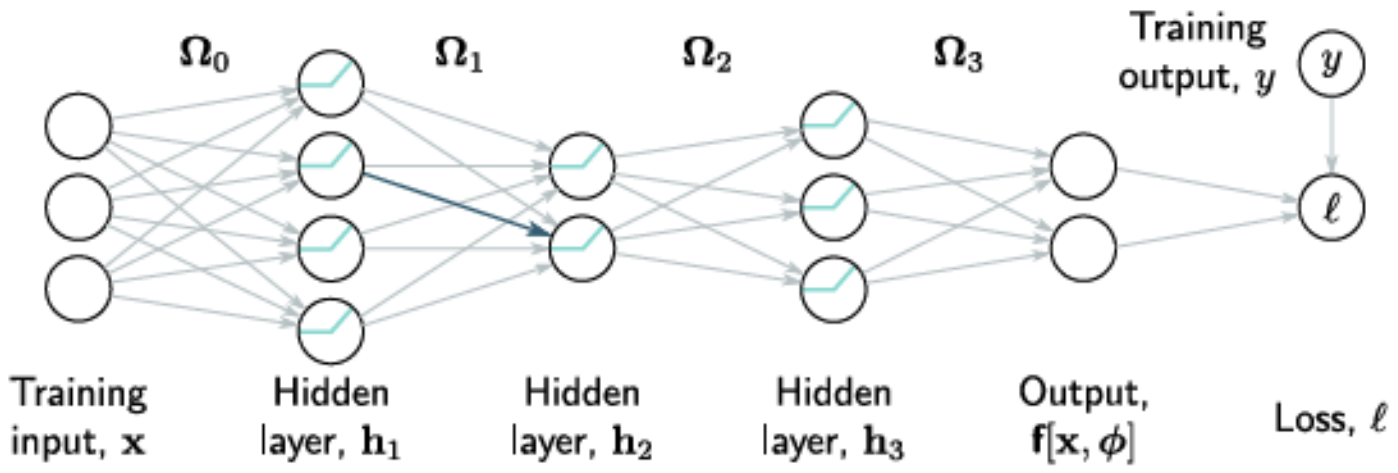

Mini-batch SGD

$$\phi_{t+1} \leftarrow \phi_t - \alpha \sum_{i \in \mathcal{B}_t} \frac{\partial l_i[\phi_t]}{\partial \phi}$$

Prince, Simon JD. *Understanding deep learning*. MIT press, 2023.

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What makes deep learning so effective?

Simplicity – Deep learning automates the most crucial step in machine learning workflow: feature engineering.

Scalability – Deep learning is highly amenable to parallelization on GPUs or more specialized machine learning hardware.

Versatility and reusability – Deep learning models can be trained on additional data without starting from scratch. They are also repurposable and thus reusable.

Prince, Simon JD. *Understanding deep learning*. MIT press, 2023.

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What deep learning has achieved so far

- Fluent and highly versatile chatbots such as ChatGPT and Gemini
- Programming assistants like GitHub Copilot
- Photorealistic image generation
- Human-level image classification
- Human-level speech transcription

Prince, Simon JD. *Understanding deep learning*. MIT press, 2023.

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What deep learning has achieved so far

- Human-level handwriting transcription and printed text transcription
- Dramatically improved machine translation
- Dramatically improved text-to-speech conversion
- Human-level autonomous driving
- Improved recommender systems

Prince, Simon JD. *Understanding deep learning*. MIT press, 2023.

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What deep learning has achieved so far

- Superhuman Go, Chess, and e-games playing

Prince, Simon JD. *Understanding deep learning*. MIT press, 2023.

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AI Hype is Real?

May 28, 2025 - Technology

Column / Behind the Curtain

Behind the Curtain: A white-collar bloodbath

Jim VandeHei, Mike Allen

Add Axios on Google

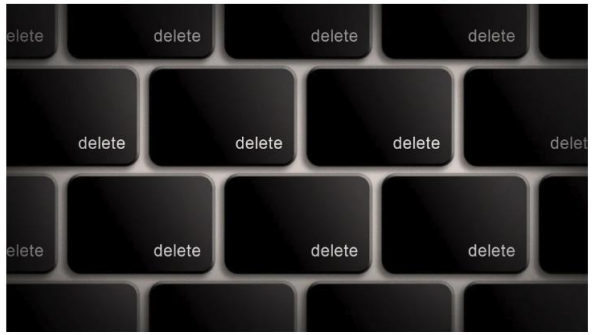

Illustration: Allie Carl/Axios

Dario Amodei — CEO of Anthropic, one of the world's most powerful creators of artificial intelligence — has a blunt, scary warning for the U.S. government and all of us:

- AI could wipe out *half* of all entry-level white-collar jobs — and spike unemployment to 10-20% in the next one to five years, Amodei told us in an interview from his San Francisco office.

WIRED

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WILL KNIGHT PARESH DAVE BUSINESS MAR 29, 2023 12:01 PM

In Sudden Alarm, Tech Doyens Call for a Pause on ChatGPT

Tech luminaries, renowned scientists, and Elon Musk warn of an "out-of-control race" to develop and deploy ever-more-powerful AI systems.

PHOTOGRAPH: N ROTTEVEEL/GETTY IMAGES

SAVE THIS STORY

AN open letter signed by hundreds of prominent artificial intelligence experts, tech entrepreneurs, and scientists calls for a pause on the development and testing of AI technologies more powerful than OpenAI's language model GPT-4 so that the risks it may pose can be properly studied

EREK BRAHNEY

ARTIFICIAL INTELLIGENCE

The great AI hype correction of 2025

Four ways to think about this year's reckoning.

by **Will Douglas Heaven**

December 15, 2025

Some disillusionment was inevitable. When OpenAI released a free web app called ChatGPT in late 2022, it changed the course of an entire industry—and several world economies. Millions of people started talking to their computers, and their computers started talking back. We were enchanted, and we expected more.

We got it. Technology companies scrambled to stay ahead, putting out rival products that outdid one another with each new release: voice, images, video. With nonstop one-upmanship, AI companies have presented each new product drop as a major breakthrough, reinforcing a widespread faith that this technology would just keep getting better. Boosters told us that progress was exponential. They posted charts plotting how far we'd come since last year's models: Look how the line goes up! Generative AI could do anything, it seemed.

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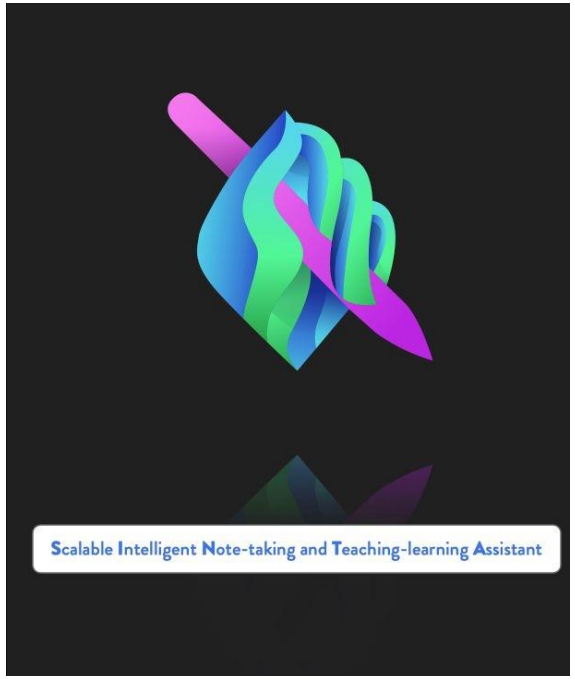
The future is bright for AI



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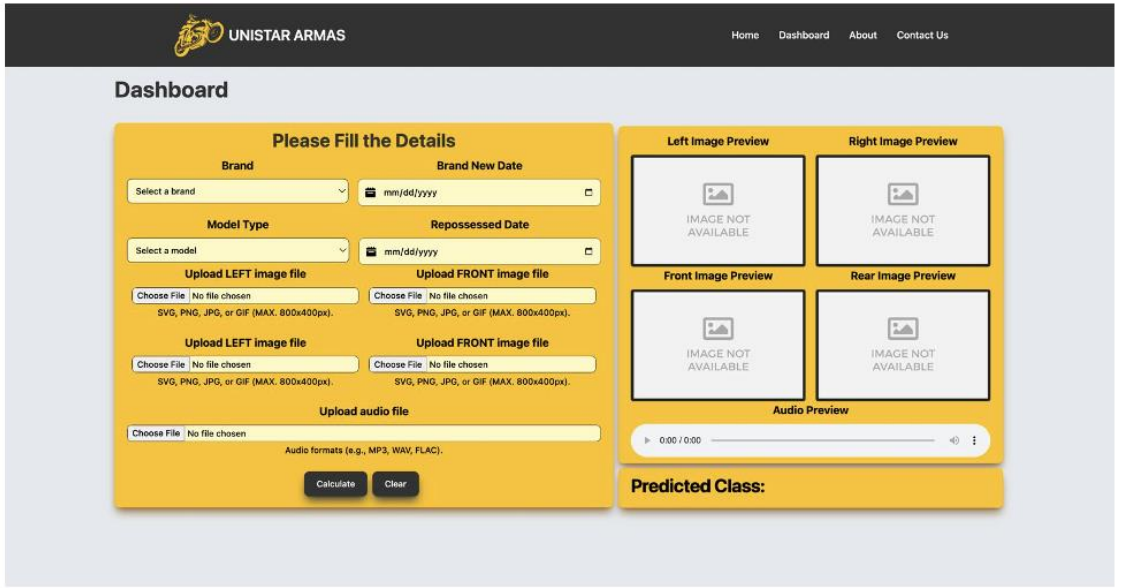
The future is bright for AI